

Shuo Guo

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Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)	Shenzhen, China
B.Sc. in Applied Mathematics	Sept 2025 – May 2029 (expected)

Research Interests

- Optimization theory, especially first-order methods, acceleration, and convergence analysis.
- Mathematical foundations of deep learning, with emphasis on optimization methods and stochastic perspectives.

Research Experience

Research Assistant, CUHK-Shenzhen	May 2026 – Jul 2026
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- Studied optimization methods used in deep learning, including adaptive gradient methods and practical training heuristics.
- Conducted preliminary experiments on small-scale language-model training and hyperparameter tuning.

Honors and Awards

- Muse Full-tuition Scholarship, CUHK-Shenzhen

Selected Academic Preparation

- **Convex Optimization:** Studied selected topics from *Lectures on Convex Optimization* by Yurii Nesterov, including convexity, gradient methods, strong convexity, complexity bounds, and accelerated first-order methods.
- **Data Science Foundations:** Studied selected topics from *Foundations of Data Science*, including singular value decomposition, random graphs, concentration phenomena, and VC dimension, through an independent course led by Prof. Zhiqian Tom Luo.
- **Deep Learning and Optimization:** Studied training dynamics, adaptive methods, and analyses of the Adam family of algorithms; read selected chapters of *Mathematical Introduction to Deep Learning: Methods, Implementations, and Theory*.
- **Stochastic Analysis:** Gained introductory exposure to stochastic-process perspectives relevant to modern analysis and machine learning.

Technical Skills

- **Programming:** Python; basic programming for mathematical, statistical, and machine-learning experiments.
- **Tools:** GitHub Copilot, Codex, and other AI-assisted coding tools.
- **Typesetting:** L^AT_EX.

Relevant Coursework

- Probability and Statistics; Linear Algebra; Calculus I and II (Honors); Computer Science; Stochastic Analysis.